

Lecture 9 - Re-thinking neurodiversity

The Social Brain: Critical Perspectives on Science, Society and Neurodiversity

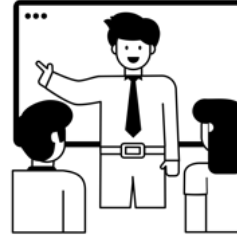
Richard Ramsey



Today

Part 1

- Re-thinking neurodiversity



Part 2

- Read articles and discuss



Overview

- Systems (not syndromes)
- Complex systems
- Dimensional frameworks
 - HiTOP
 - RDoC

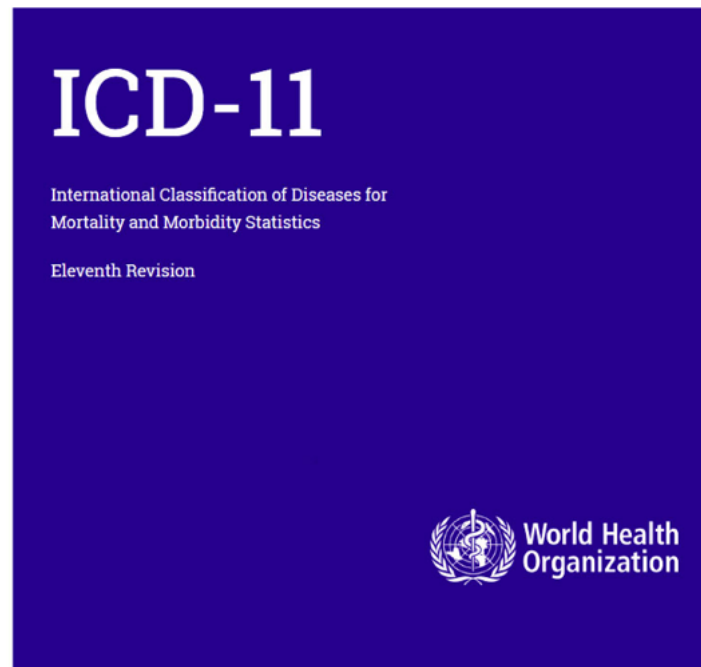
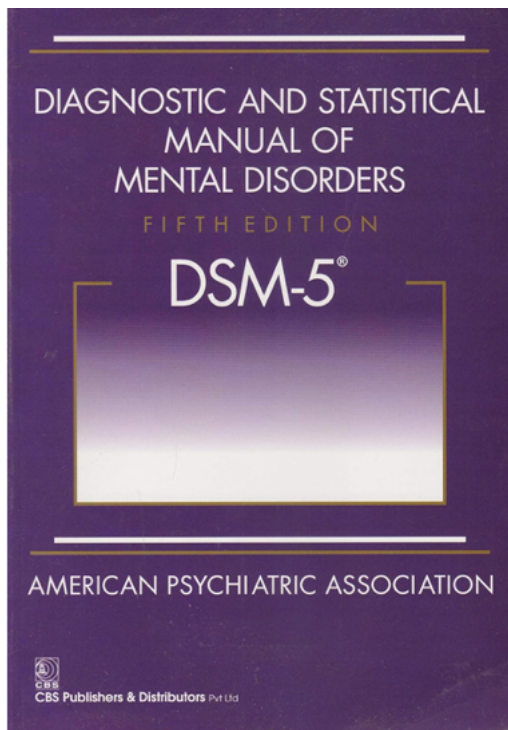


Systems (not syndromes)



Nosology

- The branch of medical science dealing with the classification of diseases.



Systems (not syndromes)

ASSOCIATION FOR
PSYCHOLOGICAL SCIENCE

Studying Mental Health Problems as Systems, Not Syndromes

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Current Directions in Psychological
Science

1–9

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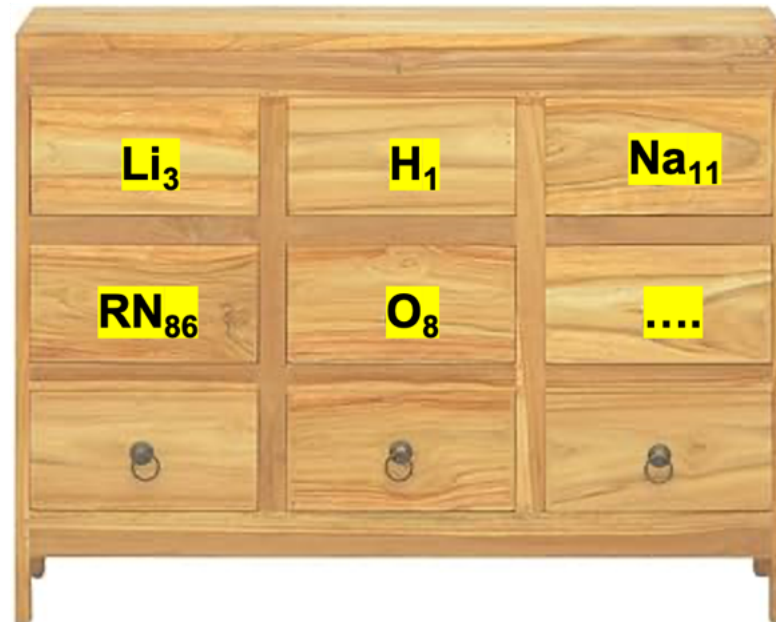
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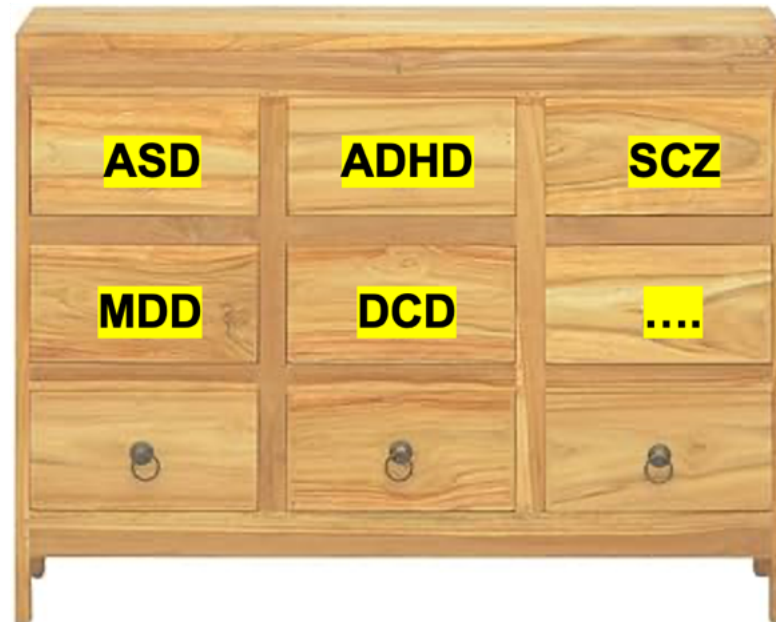
Natural kinds



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Natural kinds of disease unit?



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Historical forces, not empirical evidence

- E.g., depression DSM-5 criteria:
- 1 of the first 2 symptoms
- at least 5/9 symptoms
- for at least 14 days

How were these thresholds determined?



Diagnostic literalism

- We take diagnostic categories for more than they are, which impedes progress.

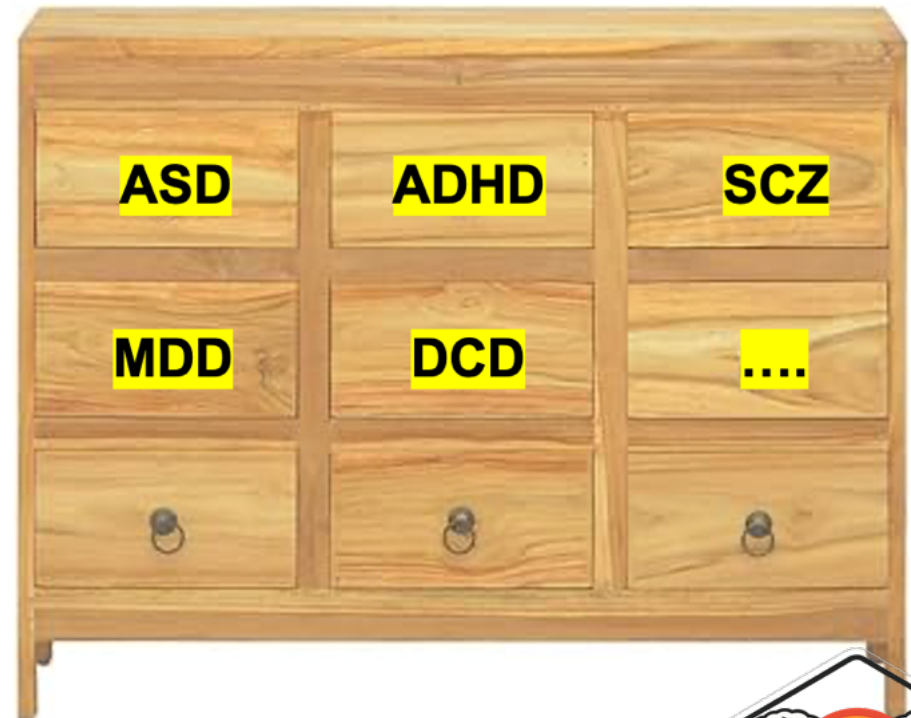
e.g., case-control studies for depression, for example, assume that depression is a neat category. Two groups: depressed vs. otherwise healthy (or at least, not depressed).

- “what are genes for depression?”
- “what are the risk factors for depression?”



Evidence does not support neat categories

- Inter-rater reliability
- Disagreement about construct definitions
- Heterogeneity
- Comorbidity of diagnostic categories
- Transdiagnostic risk factors
- Dimensions, not categories



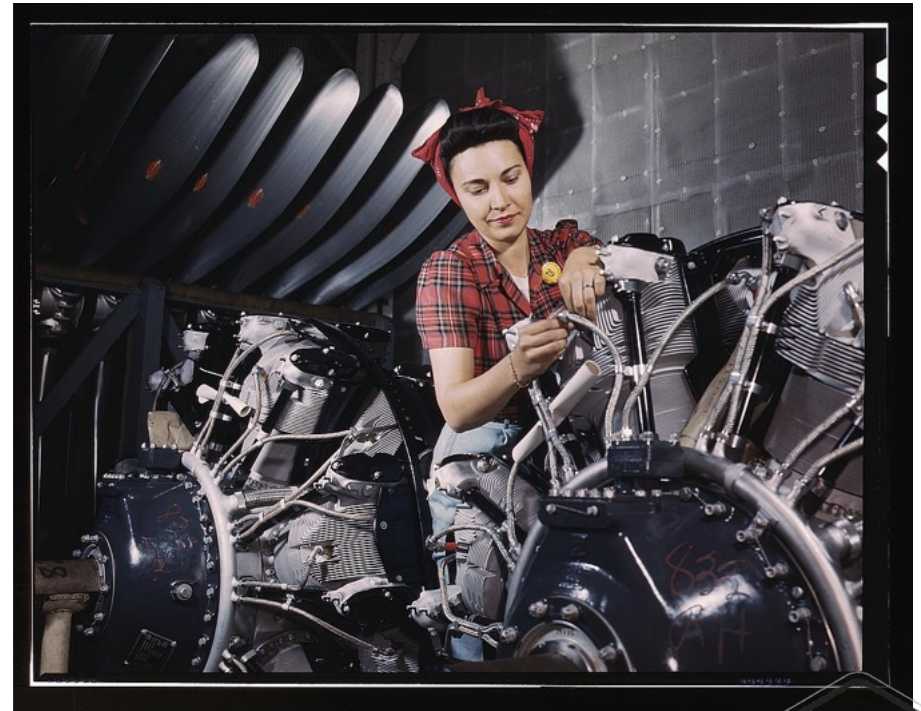
Why have categories at all?

- They can be of clinical use
- Semantic efficiency
- Public health and organisation of treatment...



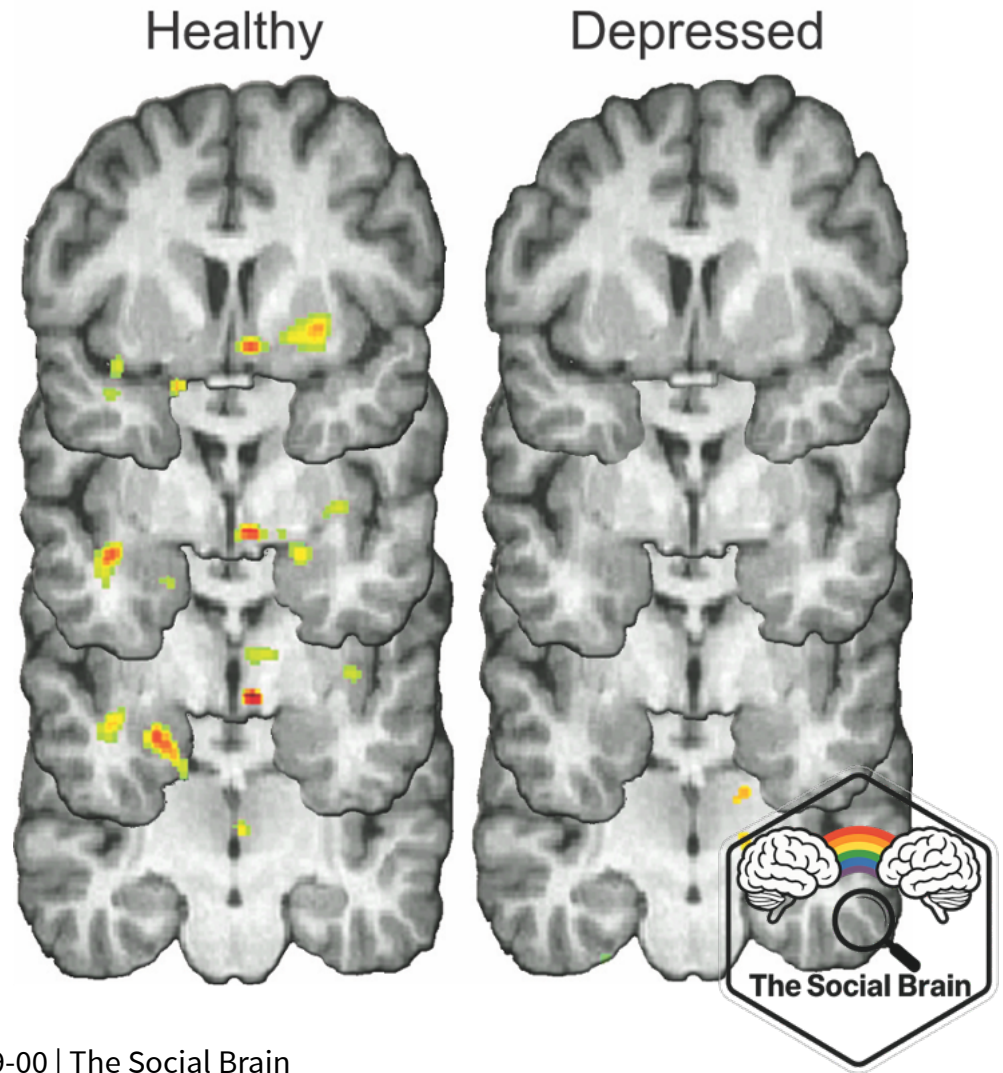
Reductionism

- Reduce things into their parts to understand the whole
- This is very useful and helpful in many different settings e.g., a car engine.
- But for complex systems, such as the stock market or weather patterns, reductionism doesn't work as well.



Reductionism in mental health

- Study depression using a DSM-5 category (which has limited validity) and only study that one category
- And focus on one level of description (i.e., a biological level)
- Primarily study one-to-one relations



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Complex systems

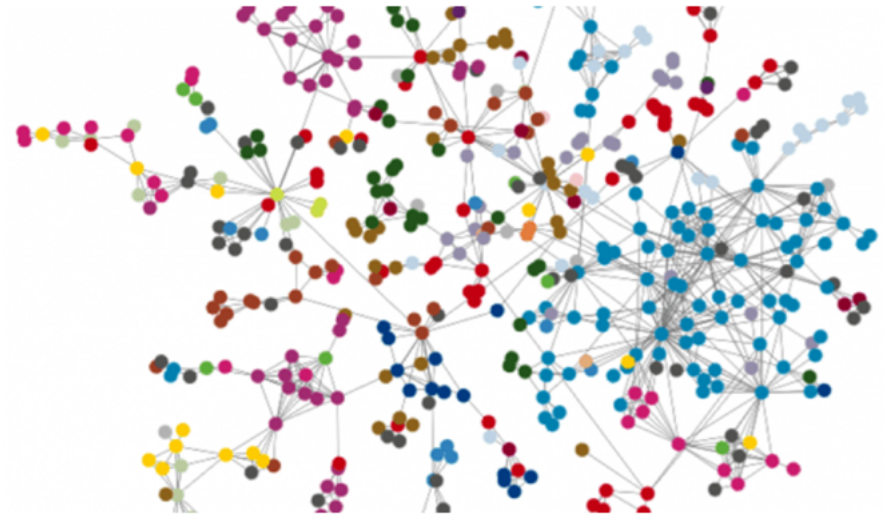


A complex systems perspective

What IS Complex Systems?



A complex systems perspective



A complex systems perspective

Lake conditions



oxygen, sunlight, fish,
pollution, etc.

Mood



personality, caffeine, sleep,
email, etc.



Mental disorders as complex systems



Interactions between:

- biological, psychological, and social features...

e.g., risk and protective factors, moods, thoughts, behaviours, predispositions, social environments.

- One-to-many or many many relations





Emergence

Flight of the Starlings: Watch This Eerie but Beautiful Phenomenon | Short Film Showcase



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Dimensional frameworks



Existing dimensional frameworks

- RDoC (NIMH, USA) e.g., Insel et al., 2010, The American Journal of Psychiatry
- HiTOP initiative (Worldwide consortium) e.g., Conway et al., 2019, Perspectives on Psychological Science



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(Conway et al. 2019, Insel et al. 2010)

HiTOP

A Hierarchical Taxonomy of Psychopathology Can Transform Mental Health Research

Christopher C. Conway¹ , Miriam K. Forbes², Kelsie T. Forbush³,

PSYCHOLOGICAL SCIENCE

Perspectives on Psychological Science
2019, Vol. 14(3) 419–436
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(Conway et al. 2019)

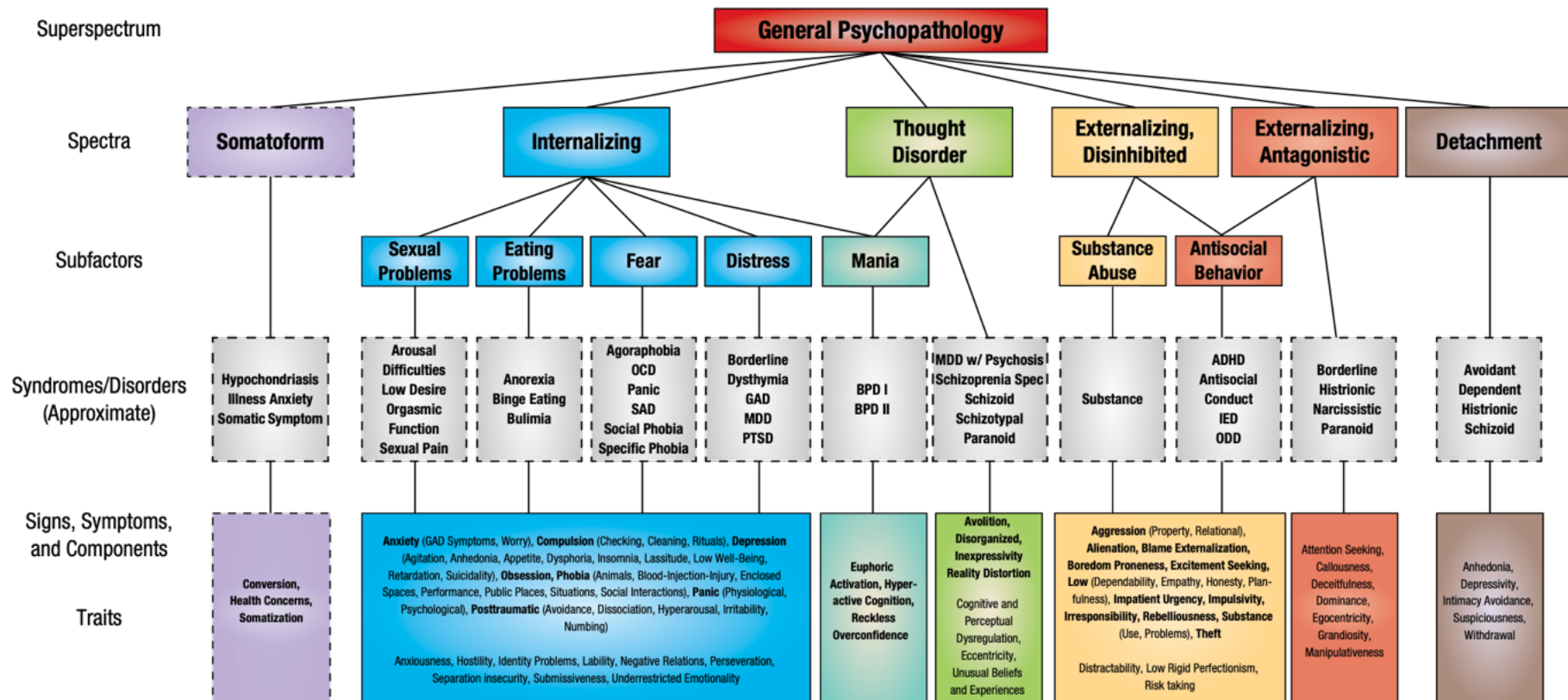
Dimensions (not categories)

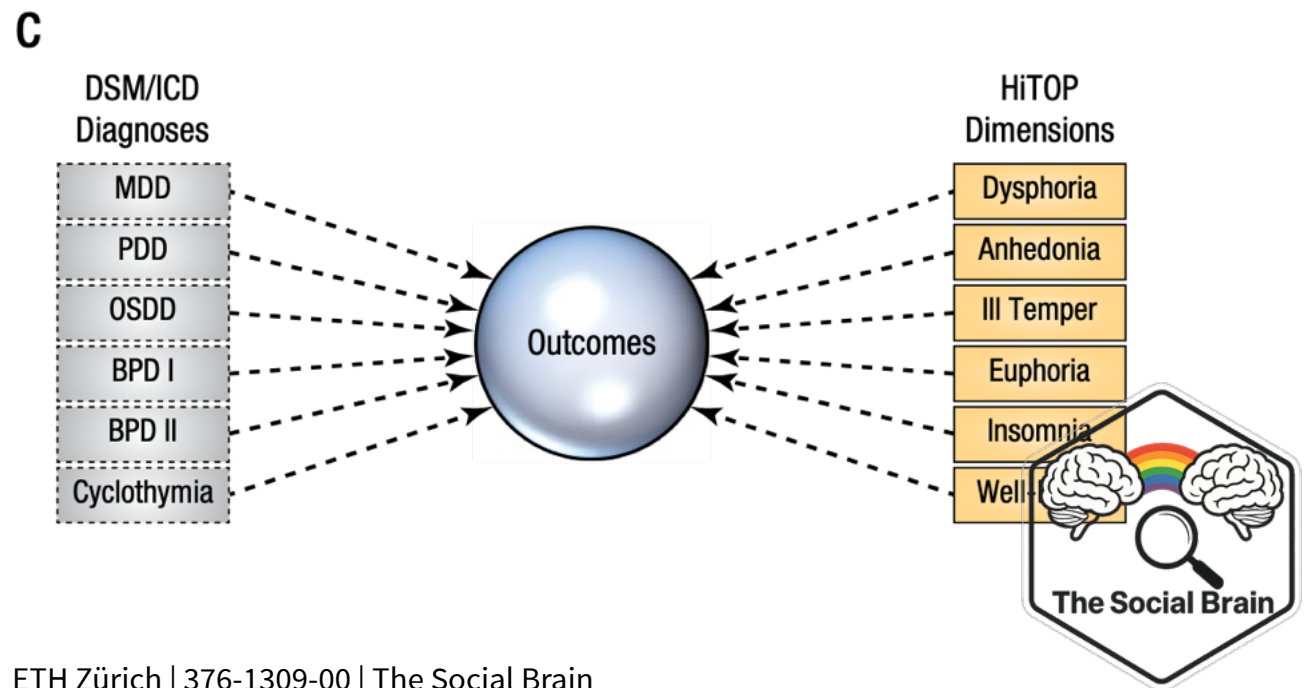
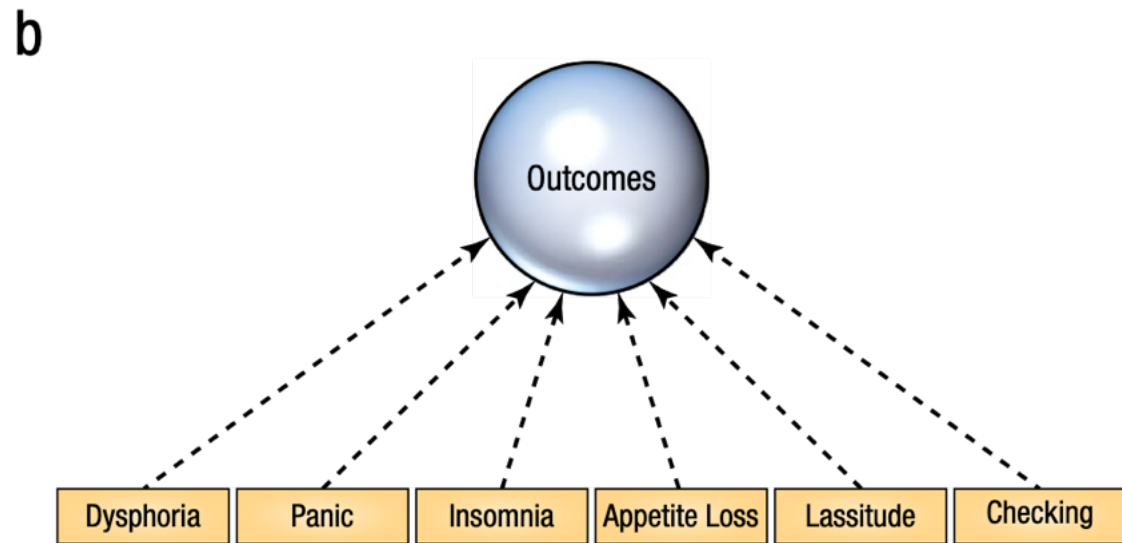
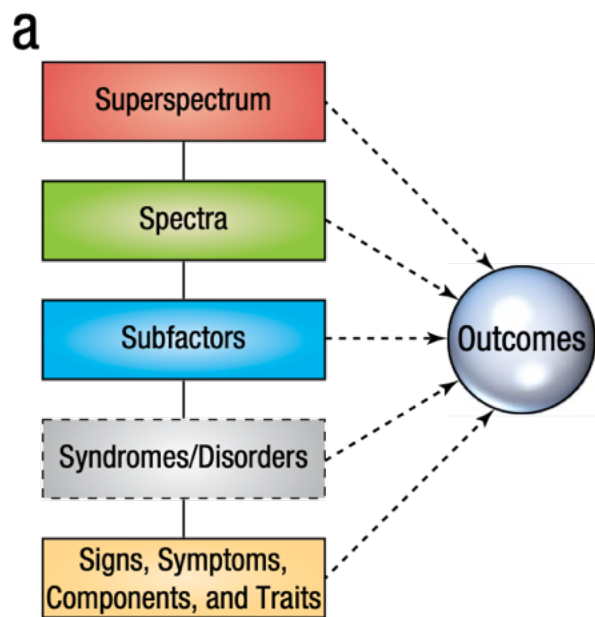
- Rather than mental health categories, we have something else...

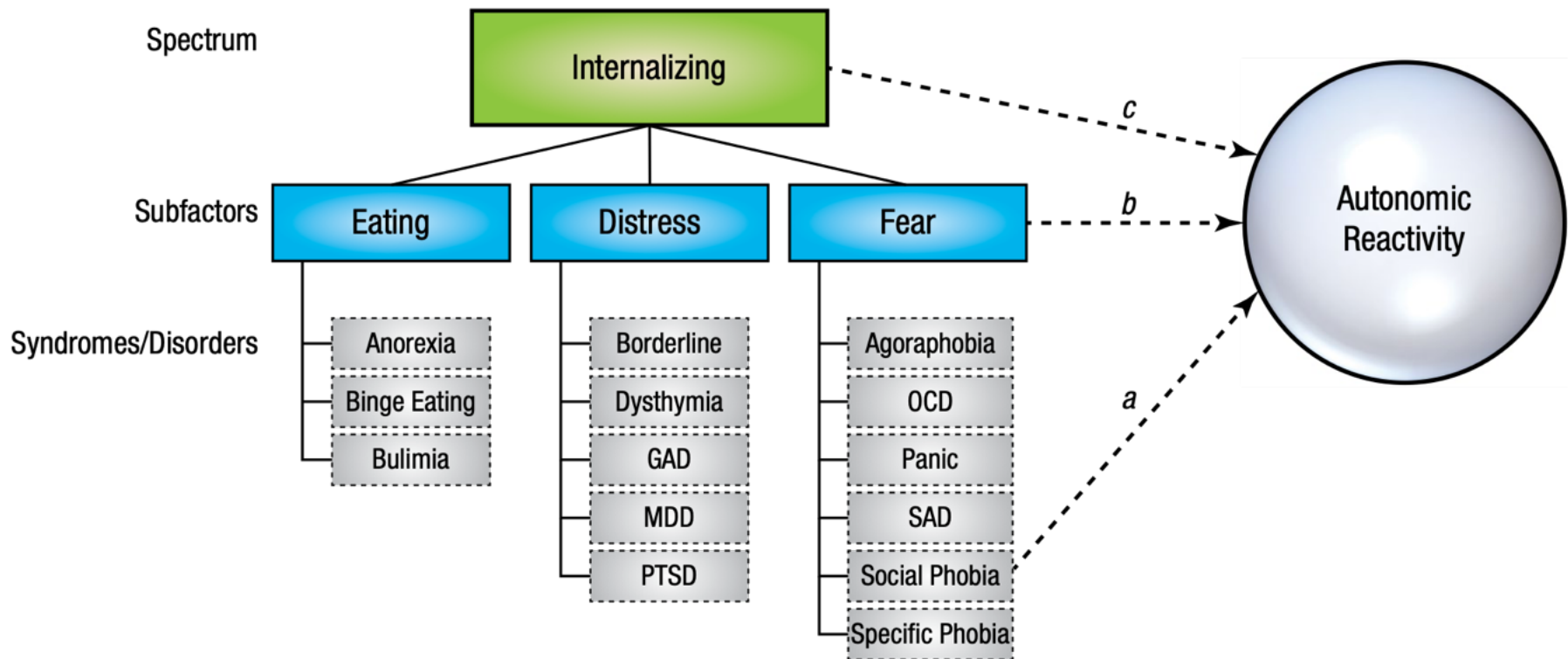
Grounded in decades of research, an alternate framework has emerged that characterizes psychopathology using empirically derived dimensions that cut across the boundaries of traditional diagnoses.

- Primary focus on evidence-based dimensions
- Dimensions evolve over time (i.e., dimensions can change if sufficient evidence is gathered)









HiTOP summary

Adopting a hierarchical and dimensional approach, makes it possible to:

- Dissect brain structure and function quantitatively and thus highlight features that
 - are common to many (or all) mental disorders
 - are specific to spectra or syndromes
 - underlie key transdiagnostic symptoms



Take a break



Part 2 - Read and discuss



Discussion material

- break into small groups (~ 5 per group)
- discuss aspects of the lecture
- discuss aspects of the journal article: <https://journals.sagepub.com/doi/10.1177/09637214221114089>



References

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- Insel, T., Cuthbert, B., Garvey, M., Heinssen, R., Pine, D. S., Quinn, K., Sanislow, C., & Wang, P. (2010). Research Domain Criteria (RDoC): Toward a New Classification Framework for Research on Mental Disorders. *American Journal of Psychiatry*, 167(7), 748–751. <https://doi.org/10.1176/appi.ajp.2010.09091379>



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